Econometrics Chapter 3: Multiple Linear Regression

(1) Recap on OLS & SLR

SLR: $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$ (Population)

Sample: $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$

Actual Reading: $Y_i = \hat{\beta}_0 + \hat{\beta}_1 X_i + \hat{\varepsilon}_i$

whereby $Y_i - \hat{Y}_i = \hat{\varepsilon}_i$ — residuals

OLS Argument: $\underset{\hat{\beta}_0^{ols}, \hat{\beta}_1^{ols}}{\operatorname{argmin}} \Sigma(Y_i - \hat{Y}_i) = \Sigma(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2$

FOC: $\hat{\beta}_0^{ols} = \bar{Y} - \hat{\beta}_1^{ols} \bar{X}$

$\hat{\beta}_1^{ols} = \dfrac{\Sigma(Y_i - \bar{Y})X_i}{\Sigma(X_i - \bar{X})^2}$

(2) Extra on OLS

a. $\hat{Y}_i^{ols}$ and $\hat{\varepsilon}_i^{ols}$ are orthogonal

Proof: $\sum \hat{Y}_i^{ols} \hat{\varepsilon}_i^{ols}$

$= \sum(\hat{\beta}_0^{ols} + \hat{\beta}_1^{ols} X_i)\hat{\varepsilon}_i^{ols}$

$= \hat{\beta}_0^{ols} \sum \hat{\varepsilon}_i + \hat{\beta}_1^{ols} \sum X_i \hat{\varepsilon}_i^{ols}$

$= 0$

b. Sample $Cov(\hat{Y}_i^{ols}, \hat{\varepsilon}_i^{ols}) = 0$

c. $\bar{Y} = \bar{\hat{Y}}^{ols} + \bar{\hat{\varepsilon}}^{ols} = \bar{\hat{Y}}^{ols}$

d. Variance Decomposition

$SST = SSE + SSR$

- Derivation: $Y_i = \hat{Y}_i^{ols} + \hat{\varepsilon}_i^{ols}$

$Y_i - \bar{Y} = \hat{Y}_i^{ols} - \bar{Y} + \hat{\varepsilon}_i^{ols}$

$(Y_i - \bar{Y})^2 = (\hat{Y}_i^{ols} - \bar{Y} + \hat{\varepsilon}_i^{ols})^2$

$= (\hat{Y}_i^{ols} - \bar{Y})^2 + 2(\hat{Y}_i^{ols} - \bar{Y})(\hat{\varepsilon}_i^{ols}) + \hat{\varepsilon}_i^2$

$\sum(Y_i - \bar{Y})^2 = \sum(\hat{Y}_i^{ols} - \bar{\hat{Y}}_i)^2 + \sum \varepsilon_i^2 + 2\sum(\hat{Y}_i^{ols} - \bar{Y}_i)(\hat{\varepsilon}_i^{ols})$

$\sum(Y_i - \bar{Y})^2 = \sum(\hat{Y}_i^{ols} - \bar{\hat{Y}}_i)^2 + \sum \varepsilon_i^2$

$SST = SSE + SSR$

uses actual observed Y_i — uses predicted $\hat{Y}_i$ from the model

e. Goodness of fit test

$R^2 = 1 - \frac{RSS}{TSS}$

$= 1 - \frac{\sum \hat{\varepsilon}_i^{2,ols}}{\sum(Y_i - \bar{Y})^2}$

- $R^2 = 1 \longrightarrow \hat{\varepsilon}_i^{ols} = 0$ for all i [Perfect Fit]
- $R^2 = 0 \longrightarrow \hat{Y}_i^{ols} = \bar{Y}$ for all i $\quad \hat{\beta}_1^{ols} = 0$

(3) Justification for OLS

a. $\hat{\beta}_1^{ols}$ is a linear estimator with the smallest variance (most efficient)

Proof:

$\hat{\beta}_1^{ols} = \sum \omega_i Y_i$ whereby $\omega_i = \frac{(X_i - \bar{x})}{\sum (X_i - \bar{x})^2}$

Allow a non-ols $\tilde{\beta}_1 = \sum (\omega_i + a_i) Y_i$

$$= \hat{\beta}_1^{ols} + \sum a_i Y_i$$

$$= \hat{\beta}_1^{ols} + \sum a_i (\beta_0 + \beta_1 X_i + \varepsilon_i)$$

$$= \hat{\beta}_1^{ols} + \beta_0 \sum a_i + \beta_1 \sum a_i X_i + \sum a_i \varepsilon_i$$

$$E(\tilde{\beta}_1 | X_1, \ldots, X_n) = \beta_1 + \beta_0 \sum a_i + \beta_1 \sum a_i X_i + \sum a_i \underbrace{(\varepsilon_i | X_1, \ldots, X_n)}_{0}$$

For $\tilde{\beta}_1$ to be unbiased,

$\sum a_i = 0$, $\sum a_i X_i = 0$

$$Var(\tilde{\beta}_1 | X_1, \ldots, X_n) = Var\left(\sum (\omega_i + a_i) Y_i\right)$$

$$= Var(\hat{\beta}_1^{ols} | X_1 \ldots) + Var(\sum a_i \varepsilon_i | X_1 \ldots)$$

$$= Var(\hat{\beta}_1^{ols} | X_1 \ldots) + \sum a_i^2 \, Var(\varepsilon_i | X_1 \ldots)$$

$$= Var(\hat{\beta}_1^{ols} | X_1 \ldots) + \underbrace{\sigma^2 \sum a_i^2}_{> 0 \text{ since } a_i^2 > 0}$$

$\therefore Var(\tilde{\beta}_1 | X_1 \ldots) > Var(\hat{\beta}_1^{ols} | X \ldots)$

Argument that $\hat{\beta}_1^{ols}$ is the most efficient with the smallest variance

b. Gauss Markov Theorem

- states that under the classical assumptions (Gauss-Markov conditions), the ols estimators of β_0 and β_1 are BLUE

- 5 Gauss-Markov Assumptions

(1) Linearity in parameters

(2) Random Sampling

(3) No Perfect Collinearity $Var(X_i) > 0$

(4) Zero Conditional Mean of errors $E(u_i | X_i) = 0$

(5) Homoskedasticity $Var(u_i | X_i) = \sigma^2$

Qns: if hetero is $E(\varepsilon | x) = 0$?

Extra: Heteroskedastic Data

- heteroskedastic data are more prevalent in economics

Allow data to be heteroskedastic

$$Var(\hat{\beta}_1^{ols} \mid X_1, \ldots, X_n) = \frac{\sum (x_i - \bar{x})^2 Var(\varepsilon_i \mid X_1 \ldots)}{(\sum (x_i - \bar{x})^2)^2}$$

Estimator for Heteroskedastic Errors

- Robust / Het Consistent / White Standard Errors

$$\hat{Var}_{HC}(\hat{\beta}_1^{ols}) = \frac{\sum (x_i - \bar{x})^2 \hat{\varepsilon}_{i,ols}^2}{(\sum (x_i - \bar{x})^2)^2} \xrightarrow{p} Var(\hat{\beta}_1^{ols})$$

- this holds despite $\hat{\varepsilon}_{i,ols}^2$ being a biased and inconsistent estimator for $Var(\varepsilon_i)$
- $\hat{Var}_{HC}(\hat{\beta}_1^{ols})$ remains consistent even under homoskedasticity
  - safer to use than $\hat{Var}(\hat{\beta}_1^{ols})$

    However, default se is likely to be better estimates of se under homoskedasticity

(4) Weighted Least Squares

Weighted Least Squares

- ols is no longer the most efficient in heteroskedasticity of error terms

Model: $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$ for all $i,j = 1,\dots,n$

whereby,

- $E(\varepsilon_i \mid X_1,\dots,X_n) = 0$
- $E(\varepsilon_i \varepsilon_j \mid X_1,\dots,X_n) = 0$
- $\Sigma(X_i - \bar{x})^2 > 0$
- $E(\varepsilon_i^2 \mid X_1,\dots,X_n) = \sigma_i^2 = \sigma^2 \eta(X_i)$ where $\eta(X_i)$ is known, assuming $X_i > 0$

WLS: Weight each observations so that weighted noise terms are no longer heteroskedastic

$$\frac{Y_i}{\sqrt{\eta(X_i)}} = \beta_0 \frac{1}{\sqrt{\eta(X_i)}} + \beta_1 \frac{X_i}{\sqrt{\eta(X_i)}} + \frac{\varepsilon_i}{\sqrt{\eta(X_i)}}$$

$$Y_i^* = \beta_0^* + \beta_1^* X_i + \varepsilon_i^*$$

- this allow ε_i^* to be homoskedastic

$$\mathrm{Var}(\varepsilon_i^* \mid X_1,\dots,X_n) = \sigma^2$$

Solving: Choosing $\hat{\beta}_0$ and $\hat{\beta}_1$ to minimize SSR

$$\Sigma(Y_i^* - \hat{\beta}_0 - \hat{\beta}_1 X_i^*)^2 = \Sigma\left(\frac{Y_i}{\sqrt{\eta(X_i)}} - \frac{1}{\sqrt{\eta(X_i)}}\hat{\beta}_0 - \frac{1}{\sqrt{\eta(X_i)}}\hat{\beta}_1 X_i\right)^2$$

$$= \frac{1}{\sqrt{\eta(X_i)}}(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2$$

$$= \Sigma w_i (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2 \quad \text{whereby } w_i = \frac{1}{\eta(X_i)}$$

Transformed Regression:

- for example, allow $\frac{Y_i}{X_i} = \beta_0 \frac{1}{X_i} + \beta_1 + \frac{\varepsilon_i}{X_i}$

$$\hat{\beta}_0^{wls} = \bar{Y}^* - \hat{\beta}_1 \bar{X}^*$$

$$\hat{\beta}_1^{wls} = \frac{\Sigma(Y_i^* - \bar{Y}^*)X_i^*}{\Sigma(X_i^* - \bar{X}^*)^2}$$

WLS Residual

- $\hat{\varepsilon}_i^{wls} = Y_i - \hat{Y}_i^{wls}$

$= Y_i - \hat{\beta}_0^{wls} - \hat{\beta}_1^{wls} X_i$

- R^2_{wls}
  - lower R^2 than OLS — outliers give less influence in WLS
    Fit of model is poorer
  - might even be negative
    indicating model fit is worse than a horizontal line model

WLS Complexity

- Form of Heteroskedasticity is known or assumed
- Form of Heteroskedasticity may have parameter that needs to be estimated

(5) Multiple Linear Regression

Multiple Linear Regression

- basic two regressor MLR model:

$Y_i = \beta_0 + \beta_1 X_i + \beta_2 Z_i + \varepsilon_i$

- OLS

$$\underset{\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2}{\operatorname{argmin}} \sum (Y_i - \hat{Y}_i)^2 = \sum (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i - \hat{\beta}_2 Z_i)^2$$

- Representative Population

$E(Y|X,Z) = \beta_0 + \beta_1 X + \beta_2 Z$ (note: including Z control for its confounding effect)

$Y = \beta_0 + \beta_1 X + \beta_2 Z + \varepsilon$

- Representative Sample

$\{X_i, Y_i, Z_i\}_{i=1}^n \overset{iid}{\sim}$ from the population

$\hat{\beta}_0^{ols}, \hat{\beta}_1^{ols}, \hat{\beta}_2^{ols}$ are unbiased and consistent estimators for $\beta_0, \beta_1, \beta_2$

Key Points of MLR:

i. Predictive relationship involving more than one variable

ii. Allows for non-linearity in $E(Y|X)$

iii. Alleviates omitted variable problems for causal interpretation of coefficients
   - in SLR, $\hat{\beta}_1^{ols}$ will be bias and inconsistent for β_1
   - Note: Confounding variables will still affect the model

iv. β_1 in MLR measures the difference in $E(Y|X,Z)$ between two members of the population holding z constant but different x
   - Discrete: $\beta_1 = E(Y|X=x+1, Z=z) - E(Y|X=x, Z=z)$
   - Continuous: $\beta_1 = \frac{\partial E(Y|X,Z)}{\partial x}$ which can be evaluated at any $X=x, Z=z$

v. β_1 and β_2 are partial regression coefficients

Building the Model: OLS Estimation of MLR Model

- Sample $\{Y_i, X_i, Z_i\}_{i=1}^n$

For any estimators, $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2$ (whether or not obtained by OLS) define:

- Fitted Values = $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i + \hat{\beta}_2 Z_i$
- Residuals: $\hat{\varepsilon}_i = Y_i - \hat{Y}_i$

$$= Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i - \hat{\beta}_2 Z_i$$

- OLS argument

$$\underset{\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2}{\operatorname{argmin}} \sum (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i - \hat{\beta}_2 Z_i)^2$$

FOC:

$$\frac{\partial SSR}{\partial \hat{\beta}_0} = -2\sum (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i - \hat{\beta}_2 Z_i) = 0$$

$$\frac{\partial SSR}{\partial \hat{\beta}_1} = -2\sum (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i - \hat{\beta}_2 Z_i) X_i = 0$$

$$\frac{\partial SSR}{\partial \hat{\beta}_2} = -2\sum (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i - \hat{\beta}_2 Z_i) Z_i = 0$$

another form of expression

$$\sum (\hat{\varepsilon}_{i,ols}) = 0$$

$$\sum (\hat{\varepsilon}_{i,ols}) X_i = 0$$

$$\sum (\hat{\varepsilon}_{i,ols}) Z_i = 0$$

- Solving for $\hat{\beta}_0^{ols}$

$$\hat{\beta}_0^{ols} = \bar{Y} - \hat{\beta}_1^{ols} \bar{X} - \hat{\beta}_2^{ols} \bar{Z}$$

- Solving for $\hat{\beta}_1^{ols}$ using auxiliary regression (Frisch-Waugh-Lovell)

Step 1: Regress X_i (Dependent) on Z_i (Independent) and collect $r_{i\,x|z}$ (How z_i affects X_i)

$$\hat{X}_i = \delta_0 + \delta_1 \hat{Z}_i$$

$$r_{i\,x|z} = X - \hat{X}_i$$

$$= X - \delta_0 - \delta_1 \hat{z}_i \quad , i = 1, \ldots, n$$

- $r_{i\,x|z}$ contains movements in X_i that are perfectly uncorrelated with z_i
- $r_{i\,x|z}$ has a sample mean of 0

Step 2: Regress Y_i on Z_i and collect $r_{i\,y|z}$

$$\hat{Y}_i = \hat{\alpha}_0 + \hat{\alpha}_1 z_i$$

$$r_{i\,y|z} = Y_i - \hat{Y}_i$$

$$= Y_i - \hat{\alpha}_0 - \hat{\alpha}_1 z_i \quad , i = 1, \ldots, n$$

- $r_{i\,y|z}$ contains movements in y_i that are perfectly uncorrelated with z_i
- $r_{i\,y|z}$ has sample mean zero

Step 3: Regress $r_{i\,y|z}$ on $r_{i\,x|z}$ using OLS

$$r_{i\,y|z} = \gamma_0 + \gamma_1 r_{i\,x|z} + u_i$$

OLS: $\sum (r_{i\,y|z} - \hat{\gamma}_0 - \hat{\gamma}_1 r_{i\,x|z}) = 0$

FOC:

- wrt $\hat{\gamma}_0^{ols}$: $\sum (r_{i\,y|z} - \hat{\gamma}_0 - \hat{\gamma}_1 r_{i\,x|z}) = 0$

$$\hat{\gamma}_0^{ols} = \overline{r_{i\,y|z}} - \hat{\gamma}_1 \overline{r_{i\,x|z}}$$

$$= 0$$

- wrt $\hat{\gamma}_1^{ols}$: $\sum (r_{i\,y|z} - \hat{\gamma}_0 - \hat{\gamma}_1 r_{i\,x|z})\, r_{i\,x|z} = 0$ ($\hat{\gamma}_0 \to 0$)

$$\sum (r_{i\,y|z} \cdot r_{i\,x|z}) - \hat{\gamma}_1 \sum (r_{i\,x|z})^2 = 0$$

$$\hat{\gamma}_1^{ols} = \frac{\sum (r_{i\,x|z} \cdot r_{i\,y|z})}{\sum (r_{i\,x|z})^2}$$

Step 4: Further Expansion on the numerator $\sum(r_{ix|z} \cdot r_{iy|z})$

$$\sum(r_{ix|z} \cdot r_{iy|z}) = \sum(r_{ix|z}(Y_i - \hat{\alpha}_0 - \hat{\alpha}_1 Z_i))$$

$$= \sum r_{ix|z} Y_i - \hat{\alpha}_0 \underbrace{\sum(r_{ix|z})}_{0} - \hat{\alpha}_1 \sum r_{ix|z} \cdot Z_i$$

0 since $r_{ix|z}$ & Z_i are uncorrelated

$$= \sum r_{ix|z} Y_i$$

$$\sum r_{ix|z} Y_i = \sum r_{ix|z}(\hat{\beta}_0 + \hat{\beta}_1 X_i + \hat{\beta}_2 Z_i + \hat{\varepsilon}_i)$$

$$= \sum r_{ix|z} \hat{\beta}_0 + \sum r_{ix|z} \hat{\beta}_1 X_i + \sum r_{ix|z} \hat{\beta}_2 Z_i + \sum r_{ix|z} \hat{\varepsilon}_i$$

$$= \hat{\beta}_0 \underbrace{\sum r_{ix|z}}_{0} + \hat{\beta}_1 \sum r_{ix|z} X_i + \hat{\beta}_2 \underbrace{\sum r_{ix|z} Z_i}_{0} + \underbrace{\sum r_{ix|z} \hat{\varepsilon}_i}_{\sum \hat{\varepsilon}_i (X_i - \hat{\gamma}_0 - \hat{\gamma}_1 Z_i)}$$

$$= \hat{\beta}_1 \sum r_{ix|z} X_i$$

$$= \sum \hat{\varepsilon}_i X_i - \hat{\gamma}_0 \sum \hat{\varepsilon}_i - \hat{\gamma}_1 \sum \hat{\varepsilon}_i Z_i$$

$$= \hat{\beta}_1 \sum r_{ix|z}(\hat{\delta}_0 + \hat{\delta}_1 Z_i + r_{ix|z}) \qquad = 0$$

$$= \hat{\beta}_1 \overset{ols}{\sum}(r_{ix|z})^2$$

Step 5: Sub Numerator $= \hat{\beta}_1^{ols} \sum(r_{ix|z})^2$

$$\hat{\gamma}_1^{ols} = \frac{\hat{\beta}_1^{ols} \sum(r_{ix|z})^2}{\sum(r_{ix|z})^2}$$

$$= \hat{\beta}_1^{ols}$$

-An Alternative Perspective to deriving MLR [OLS Decomposition]

$$\hat{Y}_i^{ols} = \hat{\beta}_1^{ols} + \hat{\beta}_{yx}^{ols} X_i$$

$$\hat{Y}_i^{ols} = \hat{\beta}_2^{ols} + \hat{\beta}_{yz}^{ols} Z_i$$

$$\hat{X}_i^{ols} = \hat{\beta}_3^{ols} + \hat{\beta}_{xz}^{ols} Z_i$$

$$\hat{Z}_i^{ols} = \hat{\beta}_4^{ols} + \hat{\beta}_{zx}^{ols} X_i$$

Then $$\hat{\beta}_1^{ols} = \frac{\hat{\beta}_{yx}^{ols} - \hat{\beta}_{yz}^{ols} \hat{\beta}_{zx}^{ols}}{1 - \hat{\beta}_{xz}^{ols} \hat{\beta}_{zx}^{ols}}$$

Assumptions for the MLR Model

- $\sum_i r_i^2 x_{i2} \neq 0$
  - X_i and Z_i cannot be perfectly correlated, if not there will be no movement in X that can explain movement in Y_i
  - Summary of requirements

$c_1 + c_2 X_i + c_3 Z_i = 0$ for all $i = 1, \ldots, n$

iff $(c_1, c_2, c_3) = 0$

Explanation: you cannot express X_i or Z_i as a linear combination of each other unless $(c_1, c_2, c_3) = 0$

(6) Properties of MLR — many of the algebraic properties are carried over from SLR

1. FOC can be written as

$$\sum \hat{\varepsilon}_i^{ols} = 0 \quad \sum \hat{\varepsilon}_i^{ols} X_i = 0 \quad \sum \hat{\varepsilon}_i^{ols} Z_i = 0$$

2. Fitted Values $\hat{Y}_i^{ols}$ and Residuals $\hat{\varepsilon}_i^{ols}$ are uncorrelated

3. $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1^{ols} \bar{X} - \hat{\beta}_2^{ols} \bar{Z}$

$(\bar{Y}, \bar{X}, \bar{Z})$ lies on the same regression function

4. $\bar{Y} = \bar{\hat{Y}}^{ols}$

$$SST = SSE + SSR$$

$$\sum (Y_i - \bar{Y})^2 = \sum (\hat{Y}_i - \bar{\hat{Y}}_i)^2 + \sum \hat{\varepsilon}_i^{2\,ols}$$

$$= \sum (\hat{Y}_i - \bar{Y}_i)^2 + \sum (Y_i - \hat{Y}_i)^2$$

$$R_2 = 1 - \frac{SSR}{SST}$$

5. OLS minimize SSR and therefore maximizes R^2

$R^2_{MLR} \geq R^2_{SLR}$

R^2 usually increases with more variables (will never decrease)

★ Never use this to decide whether to include variable

6. Adjusted R^2

- adjusted for no. of predictors

penalizing for extra predictors that may not actually improve the model's fit to the data

- Adj $R^2 = 1 - \frac{SSR/n-k-1}{SST/n-1}$

$$= 1 - \frac{SSR}{SST} \frac{n-1}{n-k-1}$$

where k is the no. of slope coefficients

7. Since $\hat{\beta}_1^{ols} = \frac{\sum r_{i\,xz} Y_i}{\sum r_{i\,xz}}$

$= \sum \omega_i Y_i$ whereby $\omega_i = \frac{\sum r_{i\,xz}}{\sum r_{i\,xz}^2}$ involves only regressor $\{X_i\}_{i=1}^n$ and $\{Z_i\}_{i=1}^n$

Weights have the following properties:

- $\sum \omega_i = 0$
- $\sum \omega_i z_i = \frac{\sum r_{i\,xz} z_i}{\sum r_{i\,xz}^2} = 0$
- $\sum \omega_i x_i = \frac{\sum r_{i\,xz} x_i}{\sum r_{i\,xz}^2} = 1$ — why?
- $\sum \omega_i^2 = \frac{1}{\sum r_{i\,xz}^2}$

8. If Sample Covariance of X_i and Z_i is zero

$\hat{\beta}_1 = \frac{\sum r_{i\,xz} Y_i}{\sum r_{i\,xz}^2} = \frac{\sum (X_i - \bar{X}) Y_i}{\sum (X_i - \bar{X})^2}$

9. Assumption $(C_1, C_2, C_3) = 0$

Implications:

- $E(\varepsilon_i | X\ldots, Z\ldots) = 0$ for all $i = 1, \ldots, n$
- $E(\varepsilon_i^2 | X\ldots, Z\ldots) = \sigma^2$ for all $i = 1, \ldots, n$
- $E(\varepsilon_i \varepsilon_j | X\ldots, Z\ldots) = 0$ for all $i = 1, \ldots, n$

10. OLS is unbiased

$\hat{\beta}_1^{ols} = \beta_1 + \sum \omega_i \varepsilon_i$

$E(\hat{\beta}_1^{ols} | X\ldots, Z\ldots) = \beta_1 + \sum \omega_i E(\varepsilon_i | X\ldots, Z\ldots)$

$= \beta_1$

$E(\hat{\beta}_1^{ols}) = \beta_1$

11. Conditional Variance of $\hat{\beta}_1^{ols}$

$$\begin{aligned}
Var(\hat{\beta}_1^{ols} \mid X\dots, Z\dots) &= Var(\beta_1 + \Sigma \omega_i \varepsilon_i \mid X\dots, Z\dots) \\
&= Var(\Sigma \omega_i \varepsilon_i \mid X\dots, Z\dots) \\
&= \Sigma \omega_i^2 \, Var(\varepsilon_i \mid X\dots, Z\dots) \\
&= \Sigma_i \omega_i^2 \, \sigma^2 \\
&= \frac{\sigma^2}{\Sigma r_i^2 x|z}
\end{aligned}$$

- $R^2_{x|z} = 1 - \frac{\Sigma r_i^2 x|z}{\Sigma (X_i - \bar{X})^2}$ (R^2 from regression of X_i on Z_i)

- Sub $\Sigma r_i^2 x|z = (1 - R^2 x|z) \Sigma (X_i - \bar{X})^2$

$$Var(\hat{\beta}_1^{ols} \mid X\dots, Z\dots) = \frac{\sigma^2}{(1 - R^2_{x|z}) \Sigma (X_i - \bar{X})^2}$$

12. Omitted Variable in SLR

- If X and Z are correlated and Z is omitted

$E(\varepsilon \mid X) \neq 0$

Regression of Y_i on X_i will give biased estimator for $\hat{\beta}_1^{ols}$

13. RS between β_2, β_1 and $Cov(X_i, Z_i)$